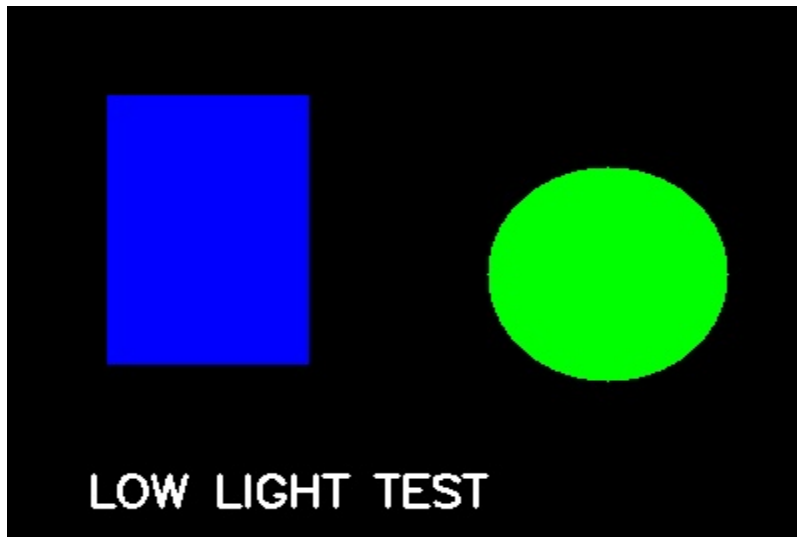


Low-Light Image Sensing and Enhancement

In low-light conditions, the number of photons reaching the camera sensor is very small. This leads to weak signal strength, increased noise, and loss of image details. To compensate, image processing techniques such as brightness enhancement and noise reduction are applied.

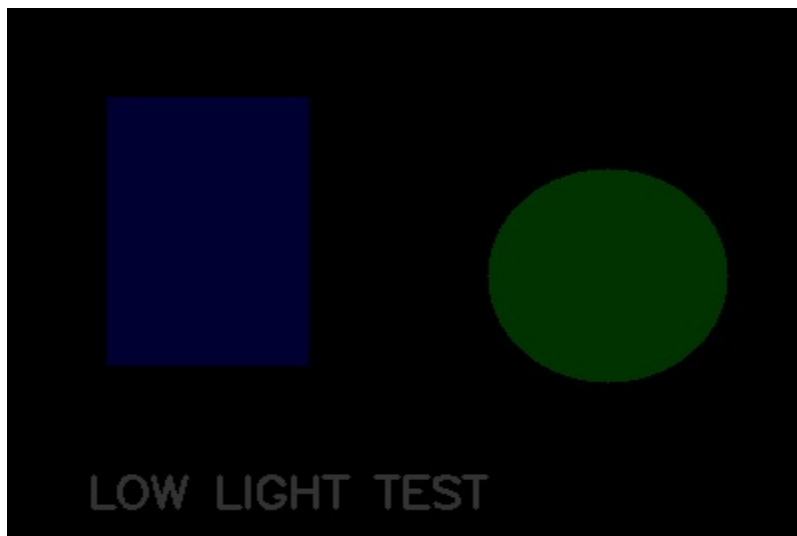
Original Image

This is the generated bright image with clear visibility and details.



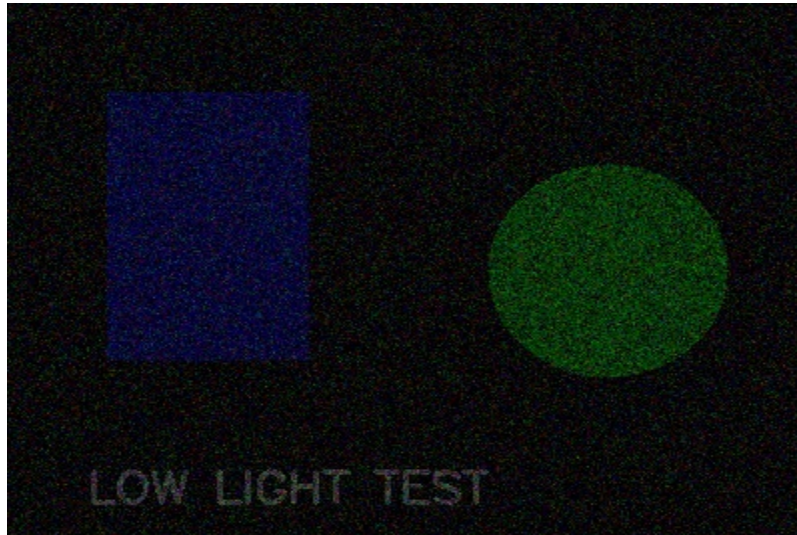
Low-Light Image

Brightness is reduced to simulate low-light conditions, representing fewer photons captured by the sensor.



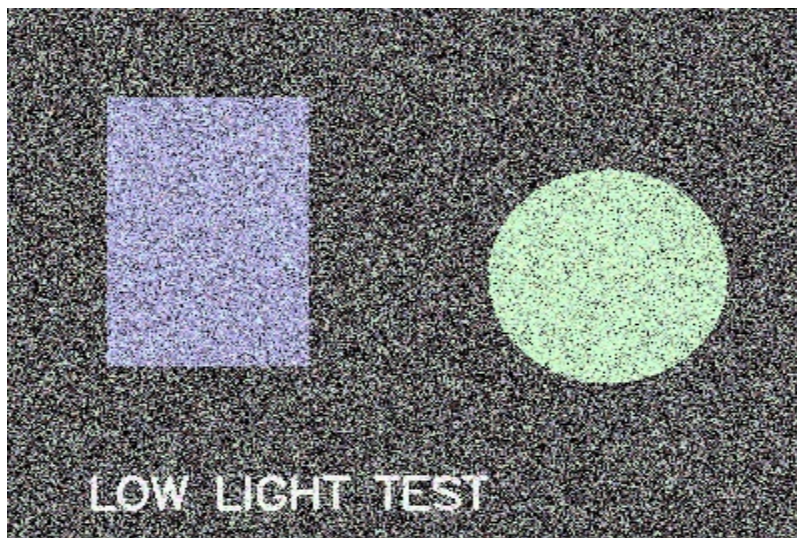
Noisy Image

Noise is added to simulate sensor noise commonly observed in low-light environments.



Enhanced Image

Histogram equalization improves brightness and enhances visibility, though some details may be lost.



Conclusion: Low-light imaging significantly affects image quality by reducing signal strength and increasing noise. Image enhancement techniques can improve visibility but cannot fully recover lost details.